Software Requirements Specification

for

Smart Parking Management System

(Vehicle Tracking, Digital Payment, and Parking Slot Booking)

Version 1.0 Initial Draft

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Revision History

Name	Date	Reason For Changes	Version
Nirupam Gupta	29/07/2026	Initial draft prepared	1.0 draft 1

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) describes the functional and non-functional requirements for Version 1.0 of the Smart Parking Management System (SPMS). The SPMS is a software-based platform that digitizes the discovery, booking, and payment of parking slots across managed parking lots, and provides real-time tracking of vehicle occupancy within those lots. This document is intended for use by the project team responsible for the design, development, testing, and deployment of the system, and by academic evaluators assessing the completeness and quality of the requirements engineering process. Unless a requirement is explicitly marked otherwise, all requirements stated in this document are considered High priority and are committed for Release 1.0.

1.2 Document Conventions

This document follows the IEEE 830-1998 recommended practice for software requirements specifications, adapted from the SRS template published by Karl E. Wiegers. Functional requirements are uniquely tagged using a mnemonic prefix identifying the parent feature followed by a sequence number (for example, PSB-3 refers to the third requirement under Parking Slot Booking). Requirement text written in the imperative form using "shall" denotes a mandatory, verifiable requirement, while "should" denotes a desirable but non-mandatory characteristic. Headings are numbered hierarchically, and all higher-level requirements are assumed to be inherited by their detailed sub-requirements unless a different priority is explicitly stated.

1.3 Intended Audience and Reading Suggestions

This SRS is intended for four categories of readers: (1) the student developer(s) who will design and implement the system; (2) the course instructor/evaluator assessing conformance to requirements engineering practice; (3) prospective end users acting as Consumers, Parking Managers, or Administrators who wish to understand the intended system behaviour; and (4) future contributors performing maintenance or extension of the system. Readers who require only a high-level understanding of the product should read Sections 1 and 2. Readers responsible for implementation or testing should additionally read Sections 3, 4, and 5 in full, along with the data dictionary in Appendix A. The document is organized to move from general context (Sections 1–2) to precise, testable detail (Sections 3–6 and appendices), consistent with IEEE 830 practice.

1.4 Product Scope

The Smart Parking Management System will allow Consumers to locate available parking slots in real time, reserve a specific slot in advance or on arrival, and pay digitally through the application, removing the inefficiency of manual, cash-based, and first-come-first-served parking operations. Parking Managers will use the system to configure parking lots and slots, monitor live occupancy through vehicle tracking, and manage on-ground bookings and exceptions. Administrators will oversee the platform as a whole: managing manager and consumer accounts, configuring pricing and lots at a portfolio level, and monitoring system health, transactions, and reports. The system directly supports the objectives of reducing vehicle idling time spent searching for parking, increasing utilisation and revenue capture for parking lot operators, and providing a transparent, cashless payment trail for every transaction.

1.5 References

1. Wiegers, Karl E. "Software Requirements Specification Template" (IEEE 830-1998 based), Process Impact, 1999.
2. IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications, Institute of Electrical and Electronics Engineers.
3. Razor Payment Gateway API Documentation (representative digital payment gateway reference), used for payment interface design.
4. Google Maps Platform Documentation, used for geolocation and slot-mapping interface reference.

2. Overall Description

2.1 Product Perspective

The Smart Parking Management System is a new, self-contained, cloud-hosted software product that replaces manual attendant-based and signage-based parking allocation with an application-driven workflow. The system is composed of a Consumer-facing mobile/web application, a Parking Manager web console, an Administrator web console, and a backend platform that exposes REST APIs consumed by all three front ends. The backend integrates with an external Digital Payment

Gateway for transaction processing and with IoT-based sensors and camera-based Automatic Number Plate Recognition (ANPR) devices installed at lot entry/exit points and individual slots for vehicle tracking. Figure 1 (Appendix B) illustrates the context of the system and its external interfaces.

The system is expected to evolve over subsequent releases to support multi-city lot networks, dynamic (demand-based) pricing, integration with municipal traffic authorities, and support for electric-vehicle charging slot reservation.

2.2 Product Functions

At a high level, the SPMS shall allow users to perform the following functions:

- Register and authenticate as a Consumer, Parking Manager, or Administrator.
- Search for, view, and book available parking slots by location, date, and time window.
- Track the real-time occupancy status of parking slots and vehicles within a lot.
- Make and receive digital payments for bookings, extensions, and penalties.
- Create, view, modify, and cancel parking slot bookings.
- Create, view, modify, and deactivate parking lots and slots (Parking Manager).
- Manage user accounts, pricing policies, and system-wide reports (Administrator).
- Receive notifications for booking confirmation, expiry, and payment events.

2.3 User Classes and Characteristics

User Class	Description
Consumer (favoured)	A member of the public who owns or drives a vehicle and wishes to locate, reserve, and pay for a parking slot. Consumers are expected to be the largest and most frequent user class, primarily interacting through a mobile application. Consumers require minimal technical expertise; the interface must therefore be simple and self-explanatory. A Consumer may register multiple vehicles and hold multiple concurrent bookings across different lots.
Parking Manager	An operator-level employee responsible for one or more physical parking lots. Parking Managers configure slot layouts and pricing for their assigned lot(s), monitor live vehicle occupancy, approve/override walk-in bookings, and handle on-site exceptions such as overstays or disputes. Parking Managers are assumed to have basic computer literacy and access to a desktop or tablet web console at the lot premises.
Administrator	A platform-level user responsible for the overall operation of the SPMS across all lots. Administrators create and deactivate Parking Manager and Consumer accounts, define system-wide business rules (e.g., cancellation policy, penalty rates), monitor transactions and generate reports, and resolve escalations that a Parking Manager cannot. Administrators are assumed to be technically proficient staff of the parking service provider.

2.4 Operating Environment

OE-1: The Consumer application shall operate on Android 10+ and iOS 14+ smartphones, and shall additionally be accessible as a responsive web application on Google Chrome, Mozilla Firefox, and Microsoft Edge (latest two major versions).

OE-2: The Parking Manager and Administrator consoles shall operate as responsive web applications on desktop browsers (Chrome, Firefox, Edge, latest two major versions) with a minimum screen resolution of 1280×720.

OE-3: The backend platform shall be hosted on a cloud infrastructure provider (e.g., AWS/GCP/Azure) running a Linux-based container runtime, and shall be horizontally scalable to accommodate multiple parking lot locations.

OE-4: The system shall interface with lot-installed IoT slot sensors and/or ANPR cameras that communicate over Wi-Fi or cellular (4G/5G) connectivity using MQTT or a comparable lightweight messaging protocol.

2.5 Design and Implementation Constraints

CO-1: The system's REST APIs shall be documented using the OpenAPI 3.0 specification.

CO-2: All persistent data shall be stored in a relational database (e.g., PostgreSQL/MySQL) with encrypted storage for personally identifiable information (PII) and payment metadata.

CO-3: All digital payment processing shall be delegated to a PCI-DSS compliant third-party payment gateway; the system shall never store raw card or bank credentials.

CO-4: All client-server communication shall occur over HTTPS (TLS 1.2 or higher).

CO-5: The system shall be developed as a modular, API-first architecture to permit independent evolution of the Consumer app, Manager console, and Admin console.

2.6 User Documentation

UD-1: The system shall provide an in-app onboarding walkthrough for first-time Consumer users covering search, booking, and payment.

UD-2: The system shall provide a searchable online help centre (FAQ and how-to articles) accessible from all three interfaces.

UD-3: The system shall provide role-specific quick-start guides (PDF) for Parking Managers and Administrators covering lot setup and account management respectively.

2.7 Assumptions and Dependencies

AS-1: It is assumed that each managed parking lot is equipped, or will be equipped prior to go-live, with functioning slot-level occupancy sensors or an ANPR camera covering all entry/exit points.

AS-2: It is assumed that Consumers have a valid mobile number and email address for account verification and notifications.

DE-1: The operation of the SPMS depends on the availability and correctness of the third-party Digital Payment Gateway for authorizing and capturing payments.

DE-2: The operation of real-time vehicle tracking depends on the availability of the IoT sensor/ANPR network operated at each parking lot; degraded sensor connectivity will degrade tracking accuracy.

DE-3: The system depends on a mapping/geolocation service (e.g., Google Maps Platform) for displaying lot locations and calculating consumer-to-lot distance.

3. External Interface Requirements

3.1 User Interfaces

UI-1: The Consumer application shall provide a map-based and list-based view of nearby parking lots, showing real-time slot availability count per lot.

UI-2: Every screen in the Parking Manager and Administrator consoles shall provide a persistent navigation menu and a contextual help icon.

UI-3: All monetary values shall be displayed in Indian Rupees (INR) with two decimal places, consistent across all three interfaces.

UI-4: The booking confirmation screen shall display a QR code that can be scanned at the lot entry gate for verification.

UI-5: All interfaces shall conform to WCAG 2.1 Level AA accessibility guidelines for colour contrast and text scaling.

3.2 Hardware Interfaces

HI-1: The system shall interface with slot-level ultrasonic/infrared occupancy sensors that report a binary occupied/vacant status per slot.

HI-2: The system shall interface with entry/exit boom-barrier controllers and ANPR cameras to log vehicle entry and exit timestamps and license plate numbers.

HI-3: The system shall interface with QR/barcode scanners installed at lot gates for booking verification.

3.3 Software Interfaces

SI-1: Digital Payment Gateway

The SPMS shall communicate with a third-party payment gateway through a programmatic REST/webhook interface for the following operations:

SI-1.1: To initiate a payment authorization request for a booking, extension, or penalty charge.

SI-1.2: To receive an asynchronous webhook confirming payment success or failure.

SI-1.3: To initiate a full or partial refund for a cancelled or disputed booking.

SI-1.4: To retrieve the settlement status of a transaction for reconciliation reporting.

SI-2: Vehicle Tracking / IoT Subsystem

SI-2.1: The SPMS shall receive real-time slot occupancy events from the IoT sensor network over MQTT and update slot status within 5 seconds of receipt.

SI-2.2: The SPMS shall receive ANPR-derived entry and exit events, including license plate number, lot ID, and timestamp, through a programmatic interface.

SI-3: Mapping and Geolocation Service

SI-3.1: The SPMS shall use the Mapping Service API to geocode lot addresses and to render lot locations on the Consumer map view.

SI-3.2: The SPMS shall use the Mapping Service API to compute estimated travel time from the Consumer's current location to a selected lot.

SI-4: Notification Service

SI-4.1: The SPMS shall use an SMS/push-notification gateway to deliver booking, payment, and reminder notifications to Consumers.

SI-4.2: The SPMS shall use an email service to send booking receipts and monthly statements.

3.4 Communications Interfaces

CI-1: The system shall send a push notification and e-mail receipt to the Consumer confirming a successful booking and payment.

CI-2: The system shall send a push notification to the Consumer 15 minutes before a time-bound booking is due to expire.

CI-3: The system shall send an SMS/push alert to the Parking Manager console when a lot reaches 95% occupancy.

CI-4: All communications interfaces shall use HTTPS/TLS for API calls and TLS-secured SMTP for e-mail delivery.

4. System Features

4.1 User Registration, Authentication, and Role Management

4.1.1 Description and Priority

Allows a person to create an account, verify their identity, log in, and access role-appropriate functionality as a Consumer, Parking Manager, or Administrator. Priority = High.

4.1.2 Stimulus/Response Sequences

Stimulus: A new user requests to register on the platform.

Response: The system collects the required profile details, verifies the mobile number/e-mail via OTP, and creates an account in the requested role (Consumer self-registers; Manager/Admin accounts are created by an Administrator).

Stimulus: A registered user requests to log in.

Response: The system validates credentials and grants access to the interface and functions corresponding to the user's role.

4.1.3 Functional Requirements

REG-1: The system shall allow a new Consumer to self-register using a mobile number or e-mail address, name, and password.

REG-2: The system shall verify a new Consumer's mobile number using a One-Time Password (OTP) before activating the account.

REG-3: The system shall allow an Administrator to create, activate, deactivate, and delete Parking Manager accounts and assign a Manager to one or more parking lots.

REG-4: The system shall allow only an Administrator to create additional Administrator accounts.

REG-5: The system shall enforce role-based access control such that a user can access only the functions permitted for their assigned role (Consumer, Parking Manager, Administrator).

REG-6: The system shall allow a Consumer to register one or more vehicles against their account, each identified by a unique license plate number and vehicle type (two-wheeler, four-wheeler, heavy vehicle).

REG-7: The system shall allow a user to reset a forgotten password through OTP-based or e-mail-link-based verification.

REG-8: The system shall lock an account for 15 minutes after five consecutive failed login attempts.

4.2 Parking Slot Search and Booking

4.2.1 Description and Priority

Allows a Consumer to search for available parking slots near a chosen location, view real-time availability, and reserve a slot for a specified date and time window, either for immediate use or in advance. Priority = High.

4.2.2 Stimulus/Response Sequences

Stimulus: Consumer requests to search for parking near a location.

Response: System queries active lots near that location and displays each lot's real-time available slot count, distance, and price.

Stimulus: Consumer selects a lot and requests to book a slot.

Response: System prompts for vehicle, date, time window, and duration, then reserves a specific slot and proceeds to payment.

Stimulus: Consumer requests to cancel a booking.

Response: If the booking status is "Confirmed" and the cancellation window has not lapsed, the system cancels the booking and releases the slot.

4.2.3 Functional Requirements

PSB-1: The system shall allow a logged-in Consumer to search for parking lots by current location, a searched address, or lot name.

PSB-2: The system shall display, for each lot in the search results, the real-time count of available slots segmented by vehicle type.

PSB-3: The system shall allow the Consumer to select a lot and view a slot-level map or list showing vacant and occupied slots.

PSB-4: The system shall allow the Consumer to specify a booking date, start time, and duration (or "until further notice" for open-ended bookings, where supported by the lot).

PSB-5: The system shall reserve a specific slot for the Consumer and place a temporary hold on that slot for 5 minutes pending payment completion.

PSB-6: If payment is not completed within the hold window, the system shall release the slot back to availability.

PSB-7: The system shall display the computed booking price, based on lot pricing rules and requested duration, before requesting payment confirmation (see BR-2).

PSB-8: Upon successful payment, the system shall assign a unique booking reference number and generate a scannable QR code for gate entry.

PSB-9: The system shall allow the Consumer to view all current and past bookings for the preceding twelve months.

PSB-10: The system shall allow the Consumer to cancel a "Confirmed" booking prior to the cancellation cutoff defined in BR-4, and shall process a refund per BR-5.

PSB-11: The system shall allow the Consumer to extend an active booking if the same slot is not reserved by another Consumer for the requested extension window.

4.3 Digital Payment Processing

4.3.1 Description and Priority

Allows a Consumer to pay for a booking, extension, or overstay penalty through supported digital payment methods, and allows the system to process refunds for eligible cancellations. Priority = High.

4.3.2 Stimulus/Response Sequences

Stimulus: Consumer confirms a booking and proceeds to payment.

Response: System calculates the payable amount and forwards a payment authorization request to the Digital Payment Gateway.

Stimulus: Digital Payment Gateway returns a payment result.

Response: If successful, the system confirms the booking and issues a digital receipt; if failed, the system notifies the Consumer and allows retry with an alternate payment method.

4.3.3 Functional Requirements

DPM-1: The system shall support payment via UPI, debit/credit card, and stored digital wallet, processed through the integrated payment gateway.

DPM-2: The system shall never store raw card numbers, CVV, or UPI PINs; only a gateway-issued payment token shall be retained for repeat-payment convenience (see SE-1).

DPM-3: The system shall display a digital receipt containing the booking reference, amount, payment method, and transaction ID immediately upon successful payment.

DPM-4: If a payment authorization fails, the system shall display the reason returned by the gateway and allow the Consumer to retry or select an alternate payment method.

DPM-5: The system shall automatically calculate and charge an overstay penalty, per BR-3, when a vehicle remains beyond the booked duration, and shall notify the Consumer of the charge.

DPM-6: The system shall process a refund to the original payment method within 5–7 business days for an eligible cancellation, per BR-5.

DPM-8: The system shall reconcile daily transaction records against the payment gateway's settlement report and flag discrepancies to the Administrator.

4.4 Real-Time Vehicle Tracking and Slot Occupancy

4.4.1 Description and Priority

Allows the system to continuously determine and display the occupancy state of every slot in a managed lot, using sensor and/or ANPR input, and to reconcile occupancy against active bookings. Priority = High.

4.4.2 Stimulus/Response Sequences

Stimulus: An IoT slot sensor or ANPR camera reports a change in occupancy for a slot or gate.

Response: System updates the slot's status (Vacant/Occupied/Reserved) in real time and reflects the change on the Consumer, Manager, and Admin views.

Stimulus: A vehicle enters a lot without a matching active booking.

Response: System flags the entry as an unregistered/walk-in vehicle and notifies the Parking Manager for manual handling.

4.4.3 Functional Requirements

VRT-1: The system shall maintain a real-time occupancy status (Vacant, Occupied, Reserved, Out-of-Service) for every slot in every managed lot.

VRT-2: The system shall update a slot's displayed status within 5 seconds of receiving a sensor or ANPR event (see SI-2.1).

VRT-3: The system shall match an ANPR-detected vehicle entry against an active booking's registered license plate number and mark the booking as "In Progress" upon a match.

VRT-4: If a vehicle enters a lot without a matching active booking, the system shall log the vehicle as a walk-in and notify the Parking Manager console.

VRT-5: The system shall log the entry and exit timestamp of every vehicle for every visit to a managed lot, for a minimum retention period of twelve months.

VRT-6: The system shall detect and flag a slot as "Out-of-Service" if its sensor reports no heartbeat for more than 10 minutes, and shall exclude it from availability search results.

4.5 Parking Lot and Slot Management (Parking Manager)

4.5.1 Description and Priority

Allows a Parking Manager to configure the physical and pricing structure of the lot(s) assigned to them, and to handle on-site booking exceptions. Priority = Medium.

4.5.2 Stimulus/Response Sequences

Stimulus: Parking Manager requests to add or edit a parking lot's slot layout.

Response: System allows the Manager to define slot count, slot type, and pricing, and persists the configuration for use by the search and booking features.

4.5.3 Functional Requirements

LOT-1: The system shall allow a Parking Manager to define a lot's slots, including slot ID, vehicle type supported, and covered/uncovered status.

LOT-2: The system shall allow a Parking Manager to set and update per-hour and per-day pricing for each slot type within their assigned lot(s), subject to the maximum pricing bounds set by the Administrator (see BR-2).

LOT-3: The system shall allow a Parking Manager to temporarily mark a slot as "Out-of-Service" for maintenance, removing it from availability search results.

LOT-4: The system shall allow a Parking Manager to create a manual walk-in booking at the lot premises for a Consumer without the mobile application, and to accept cash payment for such bookings (see BR-6).

LOT-5: The system shall allow a Parking Manager to view and export a daily occupancy and revenue report for their assigned lot(s).

LOT-6: The system shall allow a Parking Manager to override and end an active booking in the event of a dispute, subject to a mandatory reason code being recorded.

4.6 Platform Administration and Reporting (Administrator)

4.6.1 Description and Priority

Allows an Administrator to manage user accounts, define system-wide business rules, oversee all lots, and monitor platform-level analytics. Priority = Medium.

4.6.2 Stimulus/Response Sequences

Stimulus: Administrator requests to view platform-wide transaction and occupancy analytics.

Response: System aggregates data across all lots and displays configurable dashboards and exportable reports.

4.6.3 Functional Requirements

ADM-1: The system shall allow an Administrator to create, view, modify, and deactivate parking lots across the platform, and to assign one or more Parking Managers to each lot.

ADM-2: The system shall allow an Administrator to define global business rules, including the cancellation cutoff window, overstay penalty rate, and maximum permissible per-hour pricing.

ADM-3: The system shall allow an Administrator to view, search, and filter all bookings and transactions across every lot on the platform.

ADM-4: The system shall allow an Administrator to suspend a Consumer or Parking Manager account for policy violations.

ADM-5: The system shall provide the Administrator with a platform-wide dashboard showing total active bookings, revenue, occupancy rate, and flagged discrepancies (see DPM-8).

4.7 Notifications and Alerts

4.7.1 Description and Priority

Keeps Consumers, Parking Managers, and Administrators informed of relevant events through push notification, SMS, and e-mail. Priority = Medium.

4.7.2 Functional Requirements

NOT-1: The system shall send a confirmation notification to the Consumer immediately upon successful booking and payment.

NOT-2: The system shall send a reminder notification to the Consumer 15 minutes before a scheduled booking's start time and 15 minutes before its expiry.

NOT-3: The system shall notify the Parking Manager when their assigned lot reaches 95% occupancy and when a walk-in vehicle without a booking is detected.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

PE-1: The system shall support at least 5,000 concurrent Consumer sessions across the platform without degradation of response time.

PE-2: A parking slot search query shall return results within 3 seconds under normal load (up to 1,000 concurrent searches).

PE-3: The booking-to-payment confirmation round trip shall complete within 7 seconds, excluding time taken by the Consumer to enter payment details.

PE-4: Slot occupancy updates originating from a sensor/ANPR event shall be reflected on all connected clients within 5 seconds (see VRT-2).

PE-5: The system shall be able to process a minimum of 200 payment transactions per minute during peak periods.

5.2 Safety Requirements

SA-1: The system shall never allow two active bookings to be simultaneously confirmed for the same physical slot for overlapping time windows.

SA-2: In the event of a payment gateway or sensor network outage, the system shall fail safe by suspending new bookings for affected lots rather than accepting unconfirmed reservations.

SA-3: The system shall not permit an entry gate to open automatically without either a valid booking match or explicit Parking Manager authorization, to prevent unsafe/unauthorized vehicle entry.

5.3 Security Requirements

SE-1: All network transactions involving payment or personally identifiable information shall be encrypted using TLS 1.2 or higher, and sensitive data at rest shall be encrypted using AES-256 (see BR-33-equivalent CO-3).

SE-2: Users shall be authenticated via password (with OTP verification) or a supported single sign-on method before accessing any function other than public lot search.

SE-3: The system shall enforce role-based access control such that a Consumer cannot access Parking Manager or Administrator functions, and a Parking Manager cannot access lots not assigned to them.

SE-4: The system shall maintain an immutable audit log of all administrative actions, including account creation, deactivation, and business-rule changes.

SE-5: The system shall permit a Consumer to view only their own bookings, payment history, and vehicle records, not those of other Consumers.

SE-6: The system shall comply with applicable data protection regulations for the storage and processing of Consumer personal and payment data.

5.4 Software Quality Attributes

QA-1 (Availability): The system shall be available 99.5% of the time, measured monthly, excluding scheduled maintenance windows communicated at least 24 hours in advance.

QA-2 (Reliability): The system shall recover from a backend service failure and resume normal operation within 5 minutes, with no loss of confirmed booking or payment data.

QA-3 (Usability): A first-time Consumer shall be able to complete a slot search, booking, and payment within 3 minutes without external assistance, as measured during usability testing.

QA-4 (Scalability): The system architecture shall support onboarding of additional parking lots without requiring changes to existing lot configurations or downtime.

QA-5 (Maintainability): The system shall expose all inter-module communication through versioned REST APIs to allow independent updates to the Consumer, Manager, and Admin front ends.

5.5 Business Rules

BR-1: Only an Administrator may create a Parking Manager or Administrator account; Consumers may self-register.

BR-2: A Parking Manager may not set slot pricing above the platform-wide maximum defined by the Administrator (see ADM-2).

BR-3: An overstay penalty is charged at 1.5× the applicable hourly rate for every additional hour or part thereof beyond the booked duration.

BR-4: A booking may be cancelled with a full refund if cancelled at least 30 minutes before the booking start time; cancellations within 30 minutes are non-refundable.

BR-5: Eligible refunds shall be credited to the original payment method within 5–7 business days.

BR-6: Walk-in bookings created by a Parking Manager may accept cash payment; all Consumer-app bookings must be paid digitally.

BR-7: A slot marked "Out-of-Service" by a Parking Manager shall not appear in Consumer search results until reactivated.

6. Other Requirements

Internationalization: The system shall support English and Hindi language interfaces in Release 1.0, with the ability to add further regional languages in subsequent releases.

Legal: The system shall retain transaction records for a minimum of the period required by applicable financial record-keeping regulations, and shall provide Consumers a mechanism to request deletion of their personal data in accordance with applicable data protection law, subject to statutory record-retention obligations.

Reuse: Where feasible, the payment, notification, and mapping integrations shall be implemented as reusable adapter modules to support future integration with alternative third-party providers.

Appendix A: Glossary

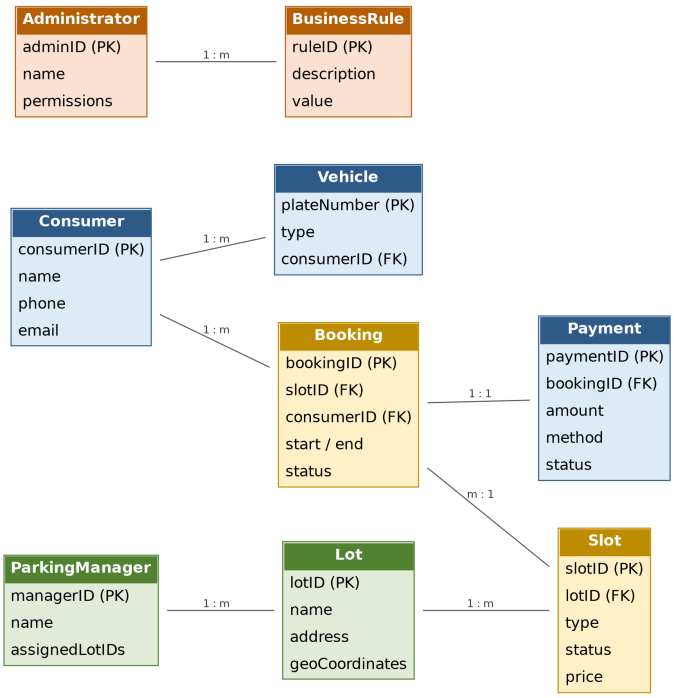
Term	Definition
ANPR	Automatic Number Plate Recognition — camera-based technology used to detect and read vehicle license plates at lot entry/exit points.
Booking	A reservation made by a Consumer for a specific parking slot, date, and time window.
Consumer	An end user who owns/drives a vehicle and uses the system to search for, book, and pay for parking.
Lot	A physical parking facility comprising one or more parking slots, managed by one or more Parking Managers.
Parking Manager	An operator-level user responsible for configuring and monitoring one or more assigned parking lots.
Slot	A single, uniquely identified parking space within a lot, capable of holding one vehicle.

Appendix B: Analysis Models

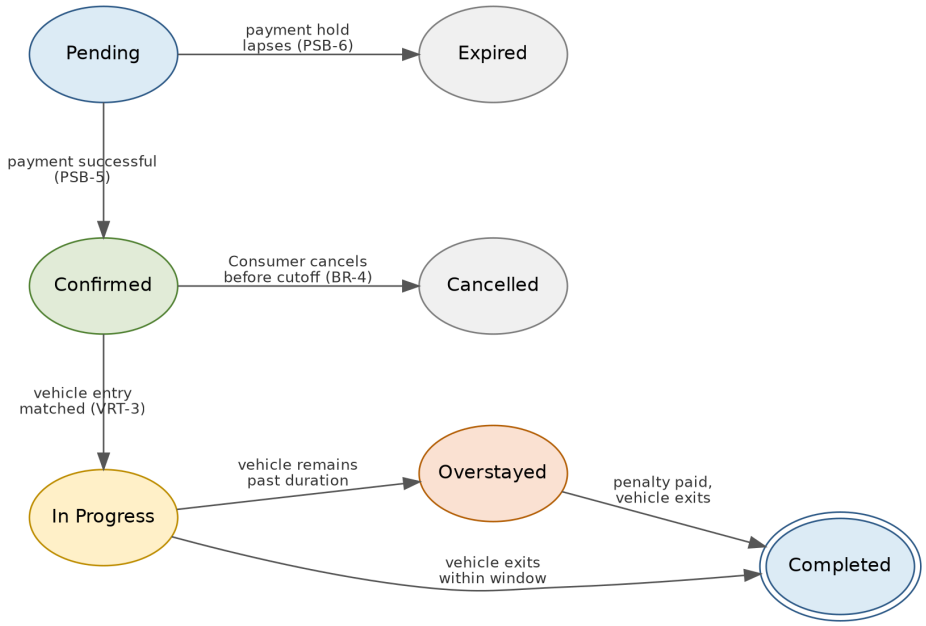
B.1 Context Diagram



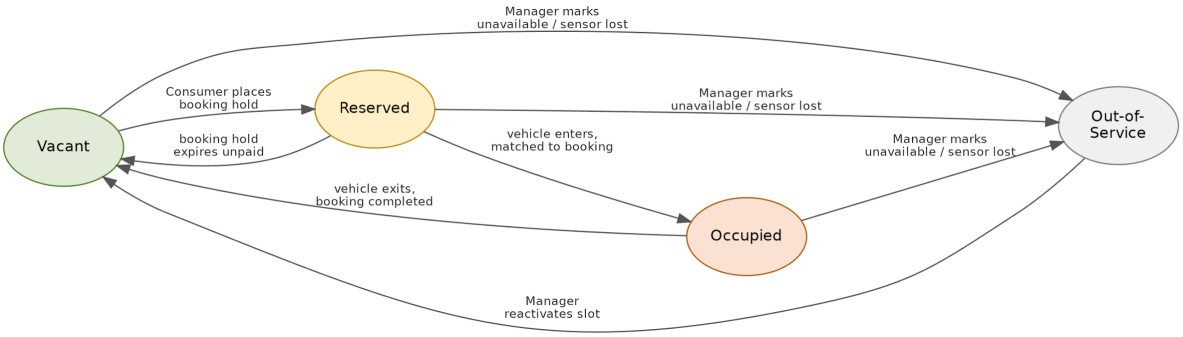
B.2 Partial Data Model (Entities and Relationships)



B.3 State-Transition Diagram — Booking Status



B.4 State-Transition Diagram — Slot Status



Appendix C: To Be Determined List

TBD #	Description
TBD-1	Final list of supported digital payment gateway providers for Release 1.0 (Section 3.3, SI-1) is pending commercial finalisation.
TBD-2	Exact overstay penalty rate and cancellation cutoff window (BR-3, BR-4) are placeholder values pending business approval; configurable via ADM-2.
TBD-3	Whether Release 1.0 will support electric-vehicle charging slot reservation, or whether this is deferred to Release 2.0.
TBD-4	Specific ANPR hardware vendor(s) to be certified for Section 3.2 hardware interfaces.

Appendix D: Requirements Traceability Matrix (Sample)

The following sample matrix illustrates how a subset of stakeholder needs identified during requirement elicitation trace to functional requirements in this SRS, supporting the Requirement Validation and Requirement Management activities of the requirements engineering process.

Stakeholder Need	Traced Requirement(s)	Elicitation Source
"I want to find a free slot without driving around"	PSB-1, PSB-2, PSB-3, VRT-1	Consumer interviews
"I don't want to carry cash to pay for parking"	DPM-1, DPM-3	Consumer survey
"I need to know which slots are actually occupied right now"	VRT-1, VRT-2, LOT-6	Parking Manager interview
"I need to control pricing across all my lots centrally"	ADM-2, LOT-2	Administrator workshop (JAD)
"I want a refund if I cancel early enough"	PSB-10, DPM-6, BR-4, BR-5	Consumer survey
"I need proof that a walk-in customer actually paid"	LOT-4, BR-6	Parking Manager interview